Capture the Flag 2: Widgets

Goal: Turn the single-card application into a simple multi-section interface.

Note from the instructor before you start:

These activities involve using your own problem solving and researching skills. We are strictly to implement NOT TO USE AI to generate the code for you. You may only use AI for RESEARCHING the syntax or usage, but NEVER THE CODE ITSELF.

Consequence: Immediate 0 score in your grade for this activity.

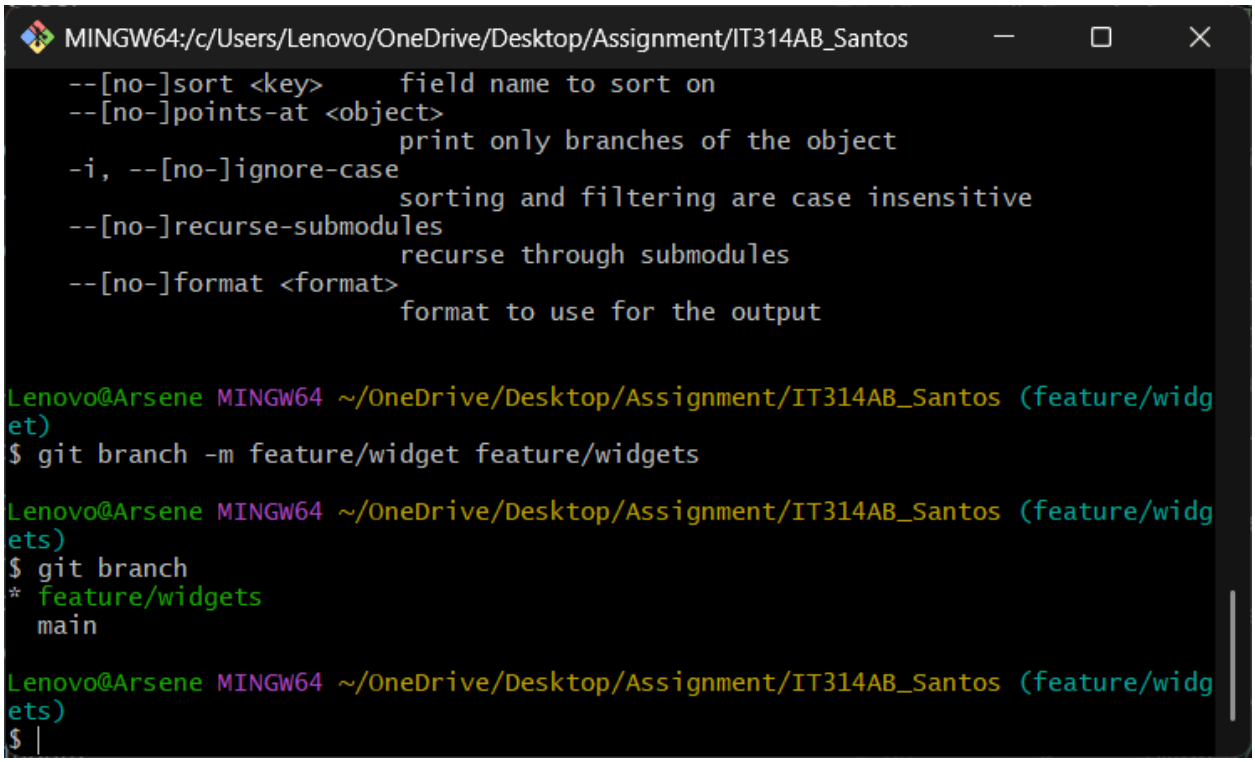
(I know what AI-generated code looks like, don't even think about it :>)

🏴 Flag 0 – Create Your Feature Branch

Objective: Create a feature branch before modifying your application. Name the branch feature/widgets

Screenshot:

- Terminal showing `feature/widgets`



🏴 Flag 1 – Break It Apart

Objective: Separate your profile information into individual `Text` widgets.

If your card currently looks like this:

```
Text(
  'Chiara Marie C. Canque\n'
  'BSIT - 3\n'
  'My First Flutter Application\n'
  'August 11, 2026',
)
```

Your task: Separate each piece of information into its own `Text` widget.

You should have something conceptually like:

Text → Name

Text → Course & Section

Text → Application Title

Text → Date

Do not worry about making it pretty yet.

Screenshot:

- Your code showing the separate `Text` widgets
- Your application

🏴 Flag 2 – Stack the Information

Objective: Arrange your individual `Text` widgets vertically.

Research how to use:

`Column()`

Place your information inside the `Column`.

Your card should now have:

Name

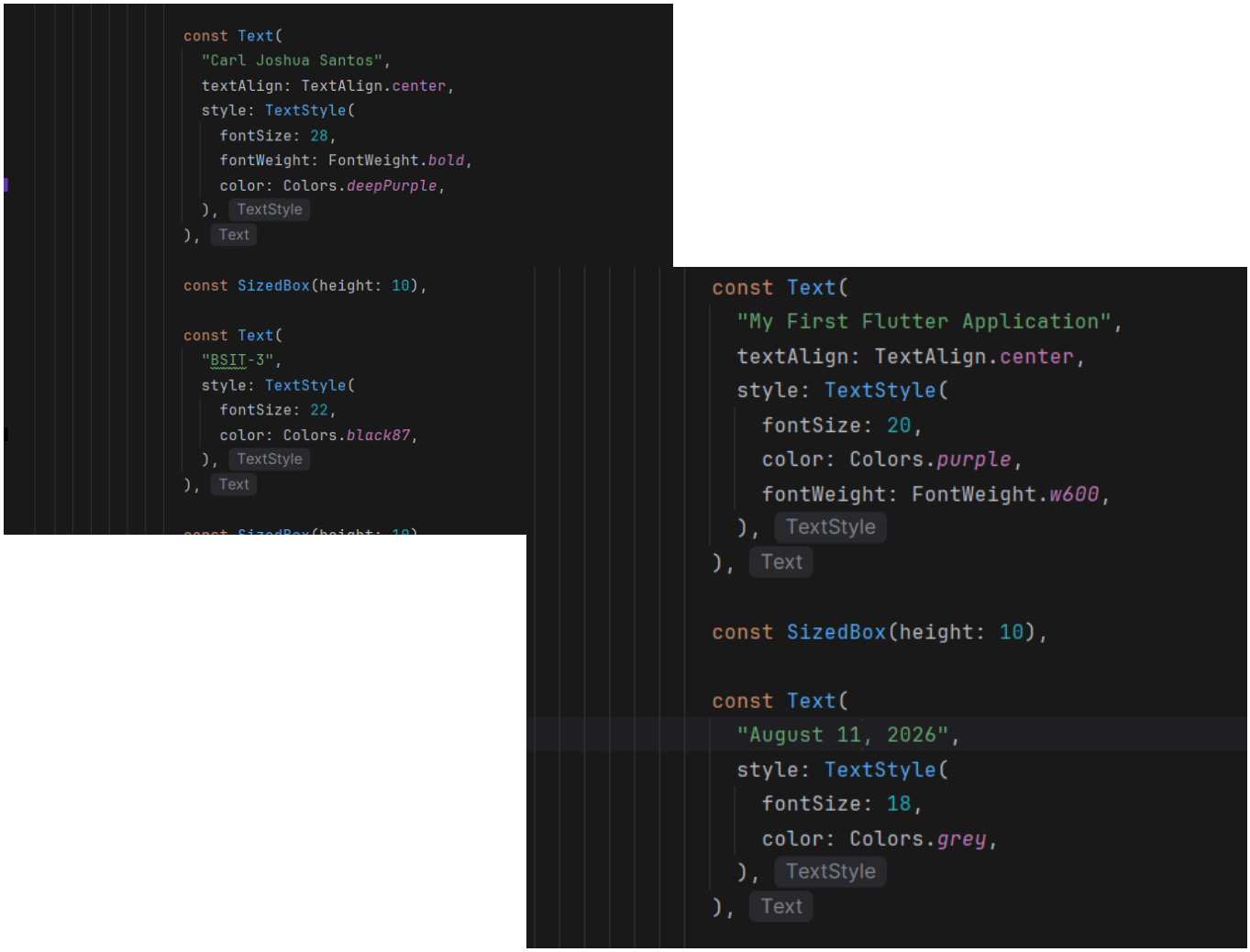
Course & Section

Application Title

Date

Screenshot:

- The `Column` in your code
- The four separate pieces of information displayed vertically




16:24
My First Flutter Application
Carl Joshua Santos
BSIT-3
My First Flutter Application
August 11, 2026
Kanade Yoisaki

🏴 Flag 3 – Add Another Element

Objective: Add another individual widget to your `Column`.

Add something that belongs to your profile.

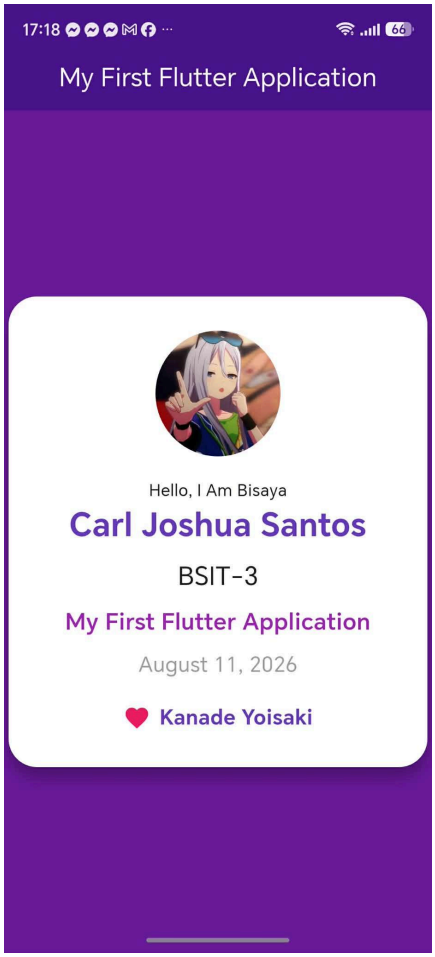
Examples:

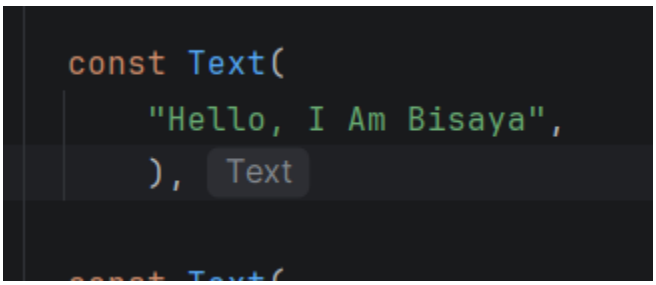
Hobby

It must be its **own widget**, not additional text inside another `Text`.

Screenshot:

- New widget in your `Column`
- Updated application





🏴 Flag 4 – Put Two Things Side-by-Side

Objective: Discover the `Row` widget.

Create a new section containing **two pieces of information beside each other**.

For example:

Age: 21 Birthdate: August 11, 2003

Research:

`Row()`

Place the `Row` inside your existing `Column`.

Your structure should now resemble:

Column

├── Text

├── Text

├── Text

├── Text

├── Text

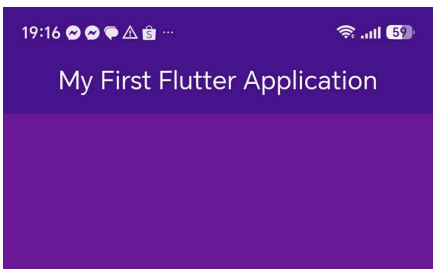
└── Row

  ├── Text

  └── Text

Screenshot:

- Your widget structure/code
- The two items displayed side-by-side




```
Row(
  mainAxisAlignment: MainAxisAlignment.center,
  children: const [
    Text(
      'Haluuuuuu',
      style: TextStyle(
        fontSize: 18,
```

🏴 Flag 5 – Give It Some Space

Objective: Improve the spacing between your widgets.

Research:

`SizedBox()`

Add appropriate spacing between your:

- Image
- Name
- Course
- Other information
- Row

Challenge: Make the card easier to read **without changing its colors or fonts yet**.

Screenshot:

- `SizedBox` usage
- Updated application

```
child: Padding(
  padding: const EdgeInsets.all(30),
  child: Column(
    mainAxisSize: MainAxisSize.min,
    children: [
      const CircleAvatar(
        radius: 55,
        backgroundImage: AssetImage('assets/Kanade-L.jpg'),
      ), CircleAvatar

      const SizedBox(height: 20),

      const Text(
          "Hello, I Am Bisaya",
          ), Text

      const SizedBox(height: 10),

      const Text(
        "Carl Joshua Santos",
        textAlign: TextAlign.center,
        style: TextStyle(
          fontSize: 28,
          fontWeight: FontWeight.bold,
          color: Colors.deepPurple,
        ), TextStyle
      ), Text
```

```
    const Text(
      "BSIT-3",
      style: TextStyle(
        fontSize: 22,
        color: Colors.black87,
      ), TextStyle
    ), Text

    const SizedBox(height: 10),

    const Text(
      "My First Flutter Application",
      textAlign: TextAlign.center,
      style: TextStyle(
        fontSize: 20,
        color: Colors.purple,
        fontWeight: FontWeight.w600,
      ), TextStyle
    ), Text

    const SizedBox(height: 10),

    const Text(
      "August 11, 2026",
      style: TextStyle(
```

```
const SizedBox(height: 10),

Row(
  mainAxisAlignment: MainAxisAlignment.center,
  children: const [

    const SizedBox(height: 10,),
```



🏴 Flag 6 – Make It Look Better

Objective: Apply styling to your individual widgets.

Now that your information is properly separated and positioned, improve the visual hierarchy.

Research how to modify:

`TextStyle()`

Apply different styling to at least **three** elements.

For example:

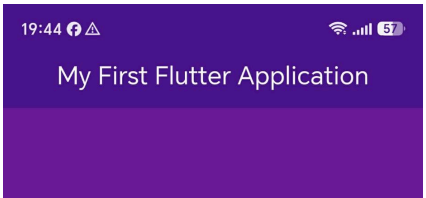
- Name → larger and bold
- Course → medium size
- Date → smaller
- Application title → italic

Screenshot:

- Your `TextStyle` changes
- Updated application

```
const SizedBox(height: 10),

const Text(
  "Carl Joshua Santos",
  textAlign: TextAlign.center,
```



```
const Text(
  "My First Flutter Application",
  textAlign: TextAlign.center,
  style: TextStyle(
    fontSize: 20,
    color: Colors.purple,
    fontWeight: FontWeight.w600,
    fontStyle: FontStyle.italic
  ), TextStyle
), Text

const SizedBox(height: 10),

const Text(
  "August 11, 2026",
  style: TextStyle(
    fontSize: 14,
    color: Colors.grey,
  ), TextStyle
), Text
```

Flag 7 – Improve the Layout

Objective: Control the alignment of your widgets.

Research the properties of `Column` and `Row`.

Experiment with:

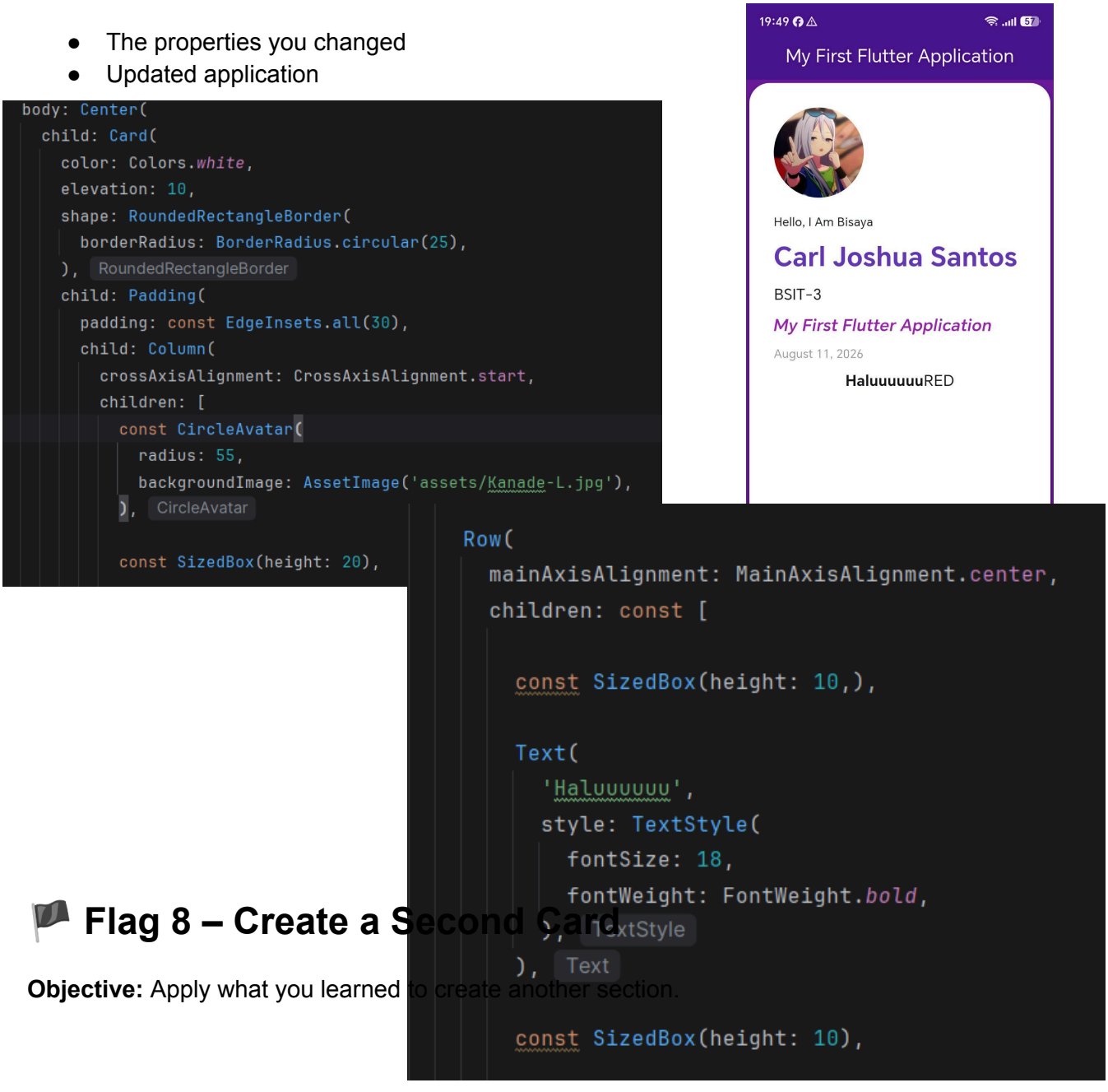
`mainAxisAlignment`

`crossAxisAlignment`

Change the alignment of your content so the card looks intentional rather than simply stacked.

Screenshot:

- The properties you changed
- Updated application



🏴 Flag 8 – Create a Second Card

Objective: Apply what you learned to create another section.

Create a second `Card` containing something different from your profile.

For example:

My Favorites

Favorite Game

Favorite Food

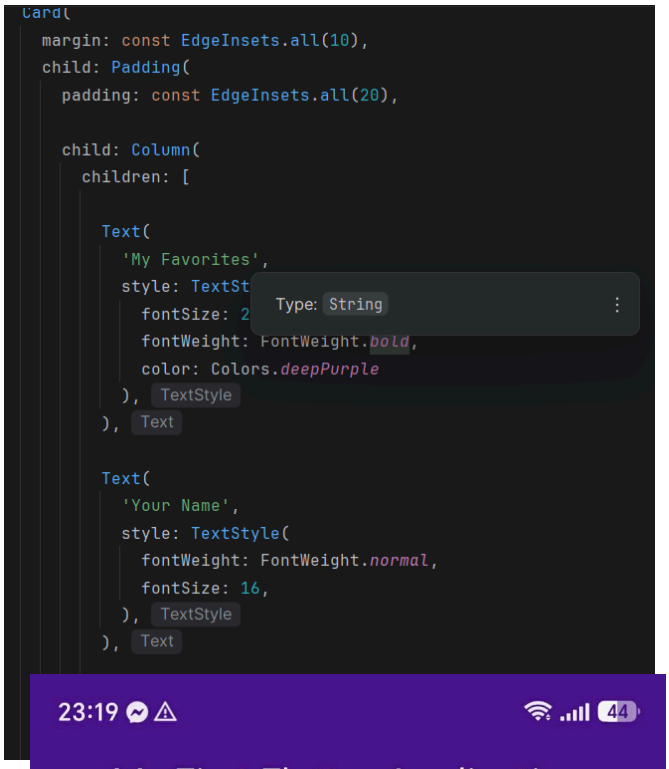
Favorite Movie

The second card should also use:

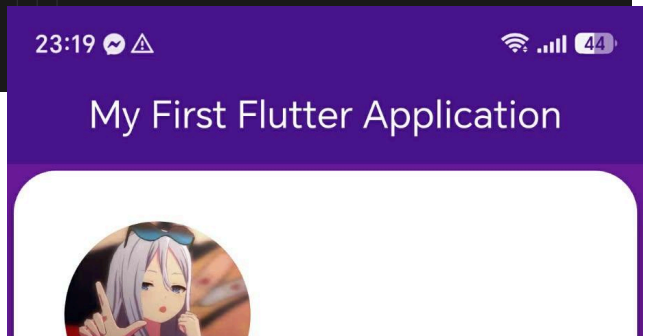
- Individual `Text` widgets
- A `Column`
- At least one `Row`
- Spacing

Screenshot:

- Code for the second card
- Both cards displayed







🏴 Flag 9 – Stack the Cards

Objective: Arrange your two cards vertically.

You now have:

Profile Card

↓

Favorites Card

Research how to place **multiple widgets inside the Flutter body**.

Use a layout widget to arrange your two cards vertically.

Screenshot:

- The widget containing both Cards
- Both cards visible in the application

```
body: SingleChildScrollView(
  child: Column(
    children: [

      Card(
        child: Padding(
          padding: const EdgeInsets.all(20),

          child: Column(
            children: [

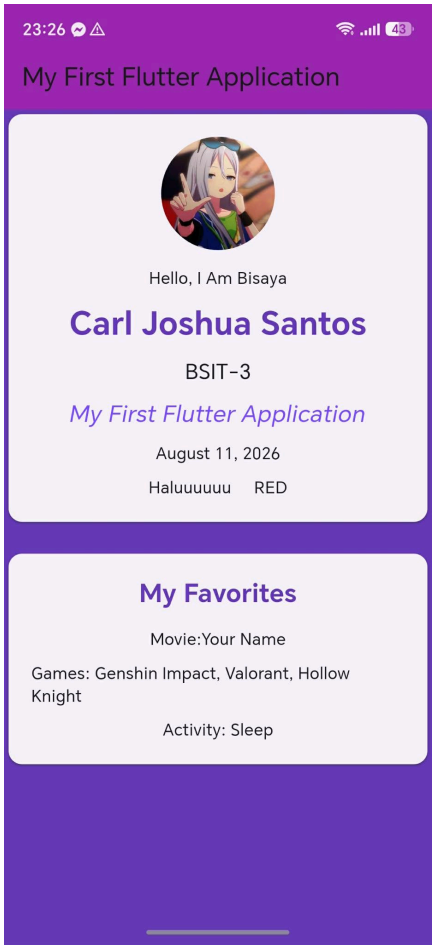
              const CircleAvatar(
                radius: 50,
                backgroundImage:
                AssetImage('assets/Kanade-L.jpg'),
              ), CircleAvatar

              const SizedBox(height: 15),

              const Text(
                "Hello, I Am Bisaya",
              ), Text

              const SizedBox(height: 10),

              const Text(
```




```
Card(
  child: Padding(
    padding: const EdgeInsets.all(20),

    child: Column(
      children: [
        const Text(
          "My Favorites",
          style: TextStyle(
            fontSize: 22,
            color: Colors.deepPurple,
            fontWeight: FontWeight.bold,
          ), TextStyle
```

🏴 Flag 10 – Widget Discovery

Objective: Discover a Flutter widget independently.

Research the Flutter widget catalog and find **one widget you haven't used yet**.

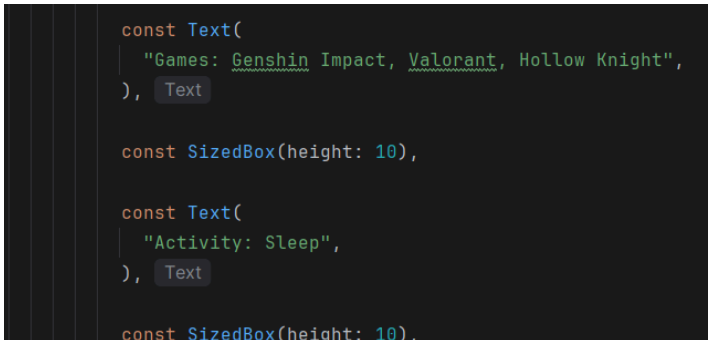
Examples:

- `Icon`
- `Divider`
- `CircleAvatar`
- `Chip`
- `ElevatedButton`

Add it somewhere appropriate in your application.

Screenshot:

- Your newly discovered widget in the code
- The widget visible in the application



```
    backgroundColor: Colors.purple,
  ), AppBar

  body: SingleChildScrollView(
    child: Column(
      children: [

        Card(
          child: Padding(
            padding: const EdgeInsets.all(20),

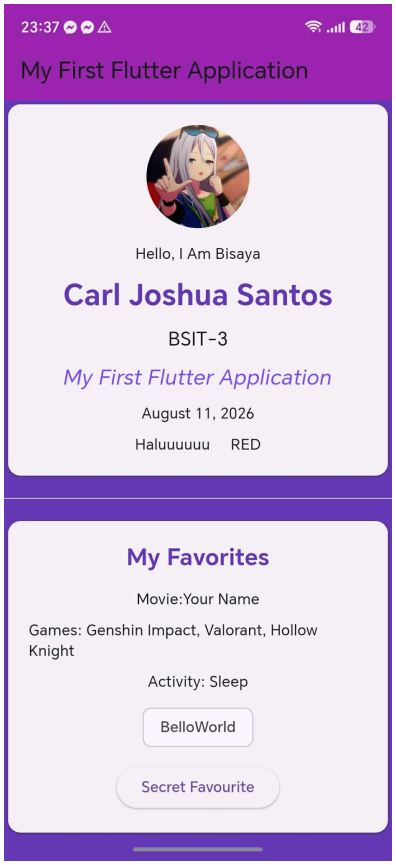
            child: Column(
              children: [

                const CircleAvatar(
                  radius: 50,
                  backgroundImage:
                  AssetImage('assets/Kanade-L.jpg'),
                ), CircleAvatar

                const SizedBox(height: 15),

                const Text(
                  "Hello, I Am Bisaya",
                ), Text
                mainAxisAlignment: MainAxisAlignment.center,
                children: const [
                  Text("Haluuuuuu"),
                  SizedBox(width: 20),
                  Text("RED"),
                ],
              ), Row
            ],
          ), Column
        ), Padding
      ), Card

      const SizedBox(height: 10),
```



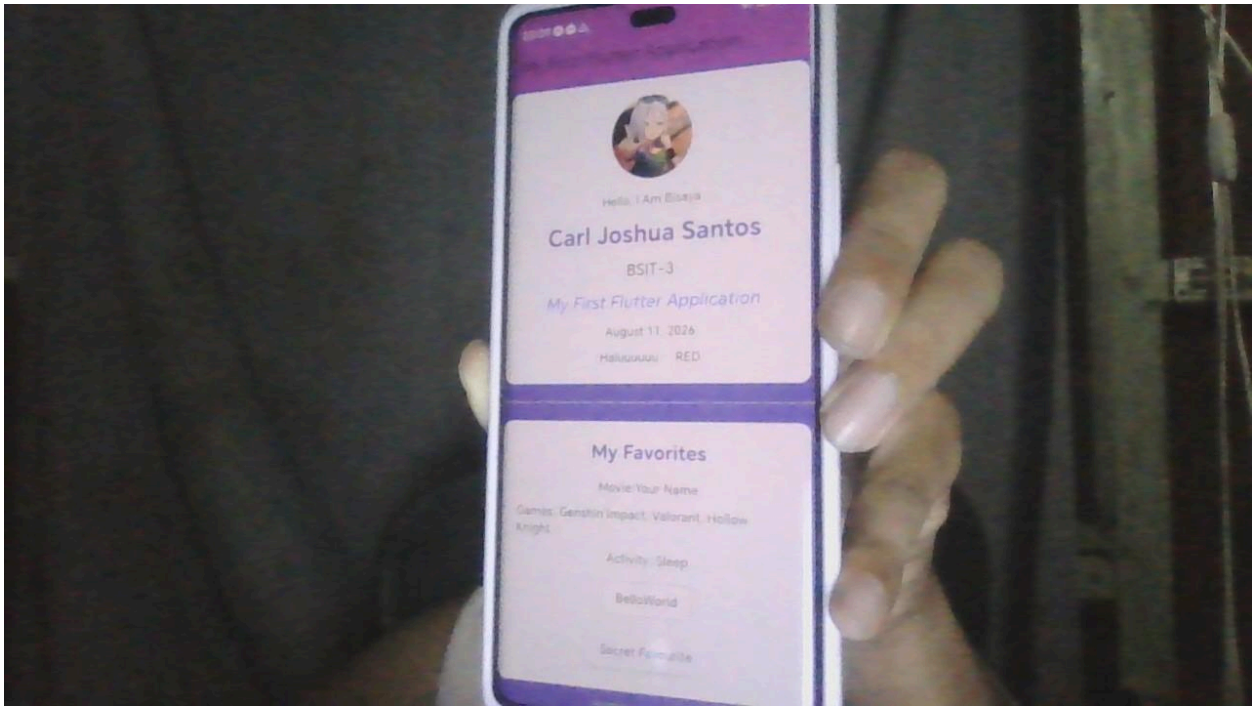
🏴 Flag 11 – Real Phone Test 📱

Objective: Run your completed application on a **physical** Android phone.

Research how to run Flutter on your phone.

Screenshot:

- Your completed application running on the physical phone (screenshot from your actual phone)



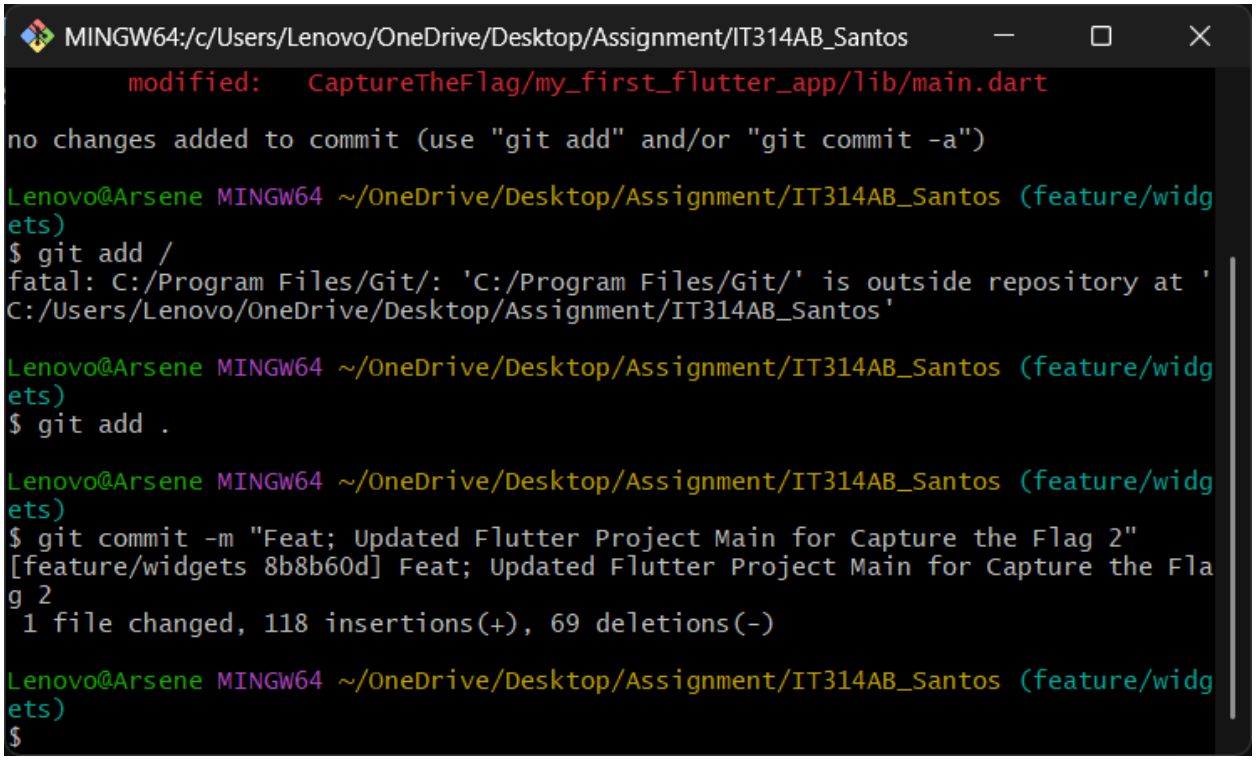
🏴 Flag 12 – Save Your Progress

Objective: Commit your completed Widget branch.

Follow proper standards of commit messages

Screenshot:

- Successful commit
- `feature/widgets` branch


MINGW64:/c/Users/Lenovo/OneDrive/Desktop/Assignment/IT314AB_Santos
modified: CaptureTheFlag/my_first_flutter_app/lib/main.dart
no changes added to commit (use "git add" and/or "git commit -a")
Lenovo@Arsene MINGW64 ~/OneDrive/Desktop/Assignment/IT314AB_Santos (feature/widgets)
$ git add /
fatal: C:/Program Files/Git/: 'C:/Program Files/Git/' is outside repository at 'C:/Users/Lenovo/OneDrive/Desktop/Assignment/IT314AB_Santos'
Lenovo@Arsene MINGW64 ~/OneDrive/Desktop/Assignment/IT314AB_Santos (feature/widgets)
$ git add .
Lenovo@Arsene MINGW64 ~/OneDrive/Desktop/Assignment/IT314AB_Santos (feature/widgets)
$ git commit -m "Feat; Updated Flutter Project Main for Capture the Flag 2"
[feature/widgets 8b8b60d] Feat; Updated Flutter Project Main for Capture the Flag 2
1 file changed, 118 insertions(+), 69 deletions(-)
Lenovo@Arsene MINGW64 ~/OneDrive/Desktop/Assignment/IT314AB_Santos (feature/widgets)
$